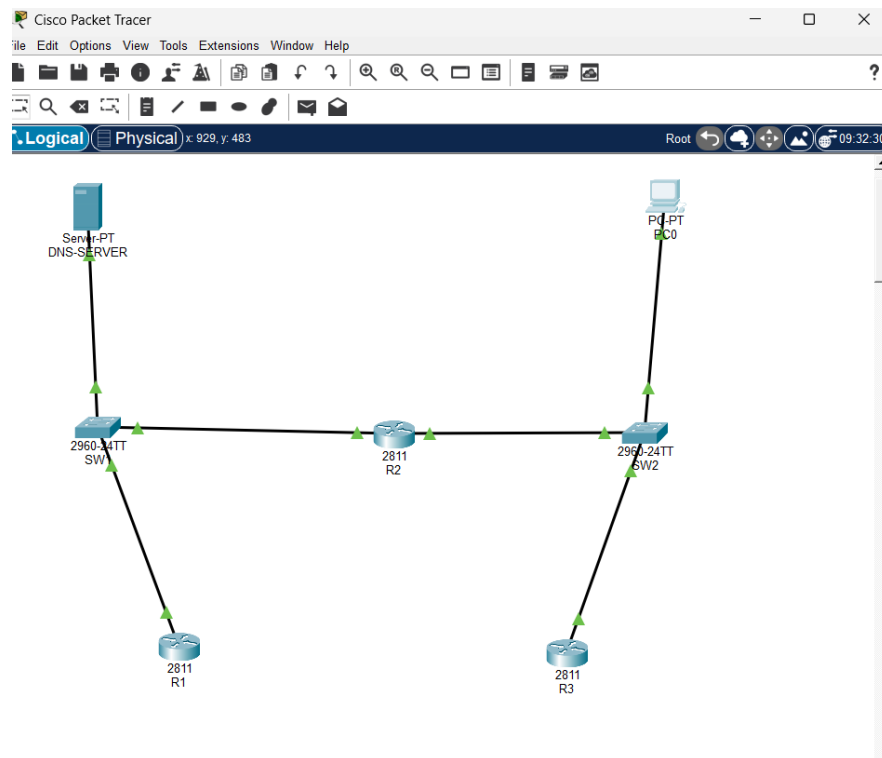


Week 4 Practical: The Cisco Troubleshooting Methodology (10 Marks)

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22) DNS resolution failed, confirms fault.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.10.10
Pinging 10.10.10.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>nslookup google.com
Server: [255.255.255.255]
Address: 255.255.255.255
DNS request timed out.
    timeout was 15000 milli seconds.
DNS request timed out.
    timeout was 15000 milli seconds.
DNS request timed out.
    timeout was 15000 milli seconds.
```

23) Layer 1: Physical

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

```
Switch>enable
Switch#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		connected	1	a-full	a-100	10/100BaseTX
Fa0/3		connected	1	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX

--More--

Layer 2: Data Link (on switch)

```
Switch#show mac address-table
      Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	0002.162b.1301	DYNAMIC	Fa0/1
1	00e0.8fa6.e702	DYNAMIC	Fa0/2

Switch#

Layer 3: Network

On PC:

```
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address. . . . .: 0006.2A55.B319
    Link-local IPv6 Address . . . . .: FE80::206:2AFF:FE55:B319
    IPv6 Address. . . . .: ::
    IPv4 Address. . . . .: 192.168.20.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0
    DHCP Servers . . . . .: 0.0.0.0
    DHCPv6 IAID. . . . .: 
    DHCPv6 Client DUID. . . . .: 00-01-00-01-5E-99-DE-91-00-06-2A-55-B3-19
    DNS Servers . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address. . . . .: 000A.4128.AAED
    Link-local IPv6 Address . . . . .: ::
```

C:\>|

On Router:

```
Press RETURN to get started!
```

```
Router>enable
Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    10.10.10.2      YES manual  up          up
FastEthernet0/1    unassigned      YES unset   up          up
Vlan1              unassigned      YES unset   administratively down down
Router#
```

Ping gateway (Failed)

```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Request timed out.
```

Layer 4: Transport + Application

```
C:\>nslookup google.com 8.8.8.8

Server: [8.8.8.8]
Address: 8.8.8.8
DNS request timed out.
        timeout was 15000 milli seconds.
DNS request timed out.
        timeout was 15000 milli seconds.
DNS request timed out.
        timeout was 15000 milli seconds.
```

24) The DNS server IP on PC0 was incorrectly configured as 10.10.10.11 instead of 10.10.10.10. The fault was at layer 3.

27) The fault was at OSI layer 3, the network layer. Based on the evidence provided on layer 1 and 2, checks passed physical links were green and MAC address table showed entries for both client and server. When I checked layer 3. Ipconfig showed the DNS server was set to 10.10.10.11 which was incorrect. The ping default gateway succeeded but ping to the DNS server failed, indicating a network layer misconfiguration.